STAT 131: Lecture Notes 1

Winter 2024

1 Classical Probability Theory

1.1 Sample spaces and events

Definition 1.1 (Experiment)**.** An *experiment* is any process where the possible *outcomes* can be identified ahead of time.

Definition 1.2 (Sample space)**.** The sample space Ω is the set of all possible outcomes of an *experiment*.

Example 1.1 (Coin tossing)**.** What is the sample space of tossing a coin twice? ♢

$\{HH, HT, TH, TT\}$

Example 1.2 (Coin tossing)**.** What is the sample space of tossing a coin three times? ♢

$\{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

Example 1.3 (Letter grades)**.** What is the sample space of a student's letter grade in a class? ♢

$\{A, B, C, D, F\}$

Example 1.4 (Human height)**.** What is the sample space of a person's height (in feet)? ♢

$(0, \infty)$

Definition 1.3 (Events)**.** An *event* is a set of possible outcomes of the experiment of the sample space.

Example 1.5 (Letter grades)**.** Events could be: (1) get an A in the class; (2) pass the course; (3) fail the course; etc.. ♢

1.2 Basic Set Theory

Definition 1.4. Set

A set is a collection of different possible outcomes.

Definition 1.5. Containment

A set A is contained in another set B if every element of the set A also belongs to the set B.

We express this using the notation $A \subset B$ or $B \supset A$

Definition 1.6. Empty Set

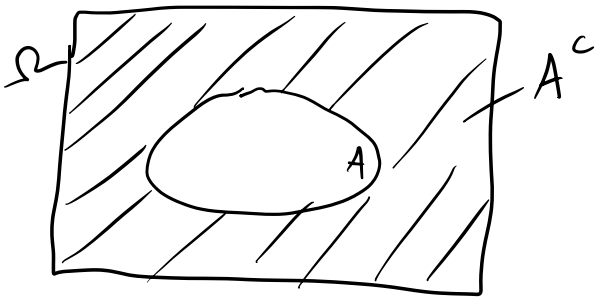
The subset of Ω that contains no elements is called the empty set or null set. It is denoted by $\emptyset$.

Definition 1.7. Complement The complement of a set A is defined to be the set that contains all elements of Ω which do not belong to A. This is denoted by A^c

Properties of complement

- $(A^C)^C = A$
- $\emptyset^C = \Omega$, $\Omega^C = \emptyset$

Example 1.6. Venn Diagram of Complement ♢



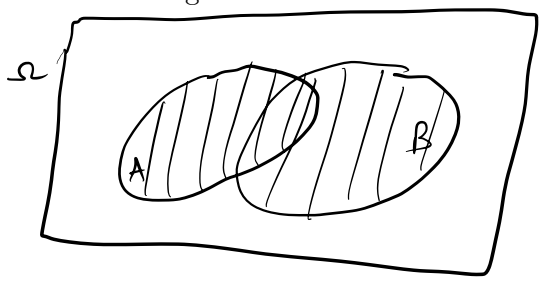
Definition 1.8. Union The union of n sets, $A_1, \dots, A_n$ is defined to be the set that contains all outcomes that belong to at least 1 of the n sets.

We denote this as $A_1 \cup A_2 \cup \cdots \cup A_n$ or $\bigcup_{i=1}^{n} A_i$

Properties of union

- $A \cup B = B \cup A$,
  $A \cup A =$ A, $A \cup A^C =$ Ω, $A \cup \emptyset =$ A, $A \cup \Omega =$ Ω
  $A \cup B =$ B, if $A \subset B$.

Example 1.7. Venn Diagram of Union ♢



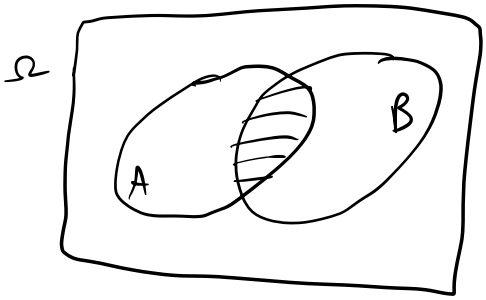
Definition 1.9. Intersection The intersection of n sets $A_1, \dots, A_n$ is defined to be the set that contains the elements that are common to all n sets.

We denote this as $A_1 \cap A_2 \cap \cdots \cap A_n$ or $\bigcap_{i=1}^{n} A_i$

Properties of intersection

- $A \cap B = B \cap A$,
  $A \cap A =$ A, $A \cap A^C =$ ∅, $A \cap \emptyset =$ ∅, $A \cap \Omega =$ A
  $A \cap B =$ A, if $A \subset B$.

Example 1.8. Venn Diagram of Intersection ♢



Definition 1.10 (Disjoint)**.** Two sets are *disjoint*, if their intersection is empty.

Theorem 1.1. *De Morgan's law*

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

Theorem 1.2. *Associative properties*

$$A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C)$$

$$A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C)$$

Theorem 1.3. *Distributive properties*

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Definition 1.11 (Partition)**.** The collection of sets $\{E_i\}_{i=1}^{\infty}$ is a partition of A, if

$$A = \cup_{i=1} E_i, \text{ and } E_i \cap E_j = \emptyset \text{ for all } i \neq j.$$

Example 1.9. The collection can be either finite or infinite. Also, partition is not unique.

- Define $E_i = \{i\}$ for $i = 1, 2, \ldots$. Then, $\{E_i\}_{i=1}^{\infty}$ is a partition of strictly positive integers.

- Define $E_1 = \{1, 2\}$ and $E_2 = \{3, 4, ...\}$. Then, B_1 and B_2 is also a partition of strictly positive integers.

♢

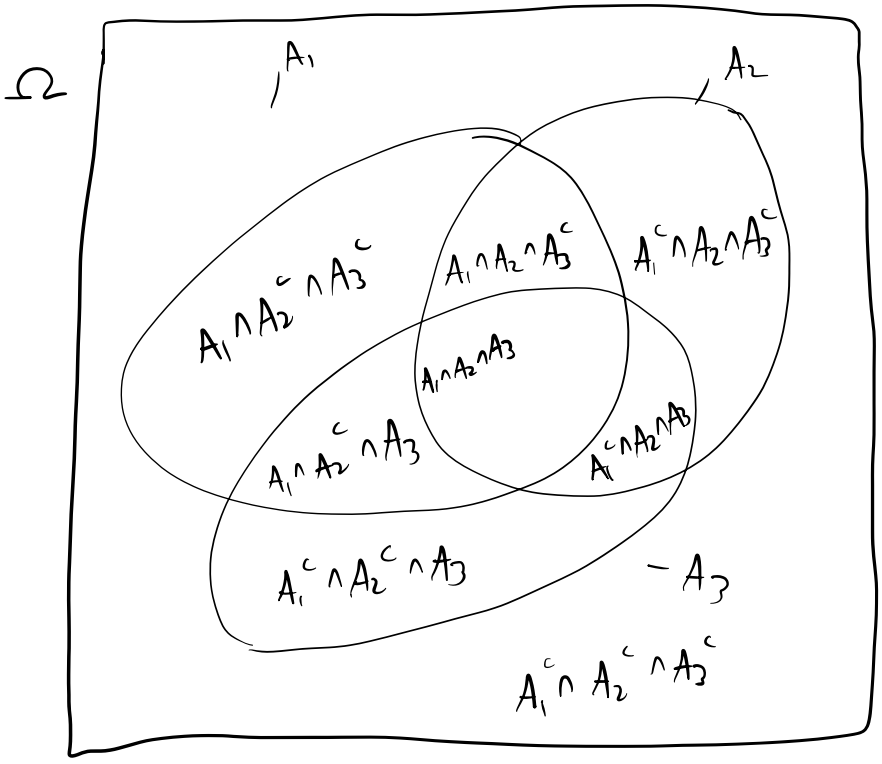
Lemma 1.4 (Partition identities)**.**

$$A = (A \cap B) \cup (A \cap B^C)$$

$$A \cup B = A \cup (B \cap A^C)$$

Note: Partitioning components *must* be disjoint.

Example 1.10. Venn Diagram of a Partition ♢



1.3 Probability

What is probability and how might we interpret it? Two ways:

- long run relative frequency.
  - Examples: Probability of rolling a 2 on a die, probability of winning a bet on red in Roulette
- subjective belief

Example 1.11. What is the chance of raining tomorrow? As we cannot repeat tomorrow, the long run relative frequency interpretation is not convincing. In contrast, the subjective belief approach is more sensible. ♢

In either case, we may use the framework in this class without inconsistency. For example, the way we'd calculate the chances of raining two days in a row would be the same, regardless of which interpretation of probability we took.
Now, towards the rigorous formulation of probability, the mathematical theory of probability is based on the following axioms.

Definition 1.12 (Axioms of probability)**.** Any probability (measure) should satisfy the following axioms.

- Axiom 1 (positivity): for every event A, $\mathbb{P}(A) \geq 0$.
- Axiom 2 (unity): $\mathbb{P}(\Omega) = 1$.
- Axiom 3 (countable additivity): For every infinite sequence of disjoint events $A_1, A_2, ...,$

$$\mathbb{P}(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

Definition 1.13. A probability measure, or simply a probability, on a sample space Ω is a specification of numbers $\mathbb{P}(A)$ for *all events* A that satisfy Axioms 1, 2, and 3.

By the definition, probability is a mapping from sets (events) to $[0, 1]$. Simply put, probability is a set-valued function, mapping sets to real numbers.

Example 1.12 (Roll one die)**.** Let's say we roll one six-sided dice, and define the probability measure to be $\mathbb{P}(A) = \frac{\text{No. of elements in event A}}{\text{No. of elements in sample space}}$. Is this a valid probability measure?

Suppose $A = \{2, 4, 6\}$

$P(A) = \frac{|A|}{|\Omega|} = \frac{3}{6} = 1/2$

$P(A) \geq 0$ for every A ✓

$P(\Omega) = \frac{|\Omega|}{|\Omega|} = 1$ ✓

$A_1 = \{1, 2\}$ $A_2 = \{3, 4\}$ $A_3 = \{5, 6\}$

$A_4 = \emptyset$ ... $A_\infty = \emptyset$

$\sum_{i=1}^{\infty} P(A_i) = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + 0 + \cdots 0$

$= 1 = P(\Omega) = P(\bigcup_{i=1}^{\infty} A_i)$ ✓ ♢

Now, let's build some basic properties of probability measures.

Lemma 1.5 (Empty set)**.** $\mathbb{P}(\emptyset) = 0$.

Proof. By Axiom 1 and Axiom 3.

$P(\Omega) = 1$ and $\Omega^c = \emptyset$

Let $A_1 = \Omega$, $A_2 = \emptyset$, $A_3 = \emptyset$. Then $\bigcup_{i=1}^{\infty} A_i = \Omega$

$P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$

$P(\Omega) = P(\Omega) + P(\emptyset) + P(\emptyset) + \cdots$

$\Rightarrow 0 = P(\emptyset) + P(\emptyset) + \cdots \Rightarrow P(\emptyset) = 0$ □

Lemma 1.6 (Finite additivity)**.** *For every finite sequence of disjoint events* $A_1, A_2, ..., A_n$*,*

$$\mathbb{P}(\cup_{i=1}^{n} A_i) = \sum_{i=1}^{n} \mathbb{P}(A_i).$$

Proof. By Axiom 3 and $\mathbb{P}(\emptyset) = 0$.

Let $A_1 = A_1$, $A_2 = A_2$, ..., $A_n = A_n$ and $A_i = \emptyset$ for $i > n$

Then $\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{n} A_i$

and $P(\bigcup_{i=1}^{n} A_i) = P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$

$= \sum_{i=1}^{n} P(A_i) + \sum_{i>n} P(A_i)$

$= \sum_{i=1}^{n} P(A_i) + 0$

$= \sum_{i=1}^{n} P(A_i)$ □

The following properties can be shown via the Axioms.

Theorem 1.7. *Complement:* $\mathbb{P}(A^C) = 1 - \mathbb{P}(A)$.

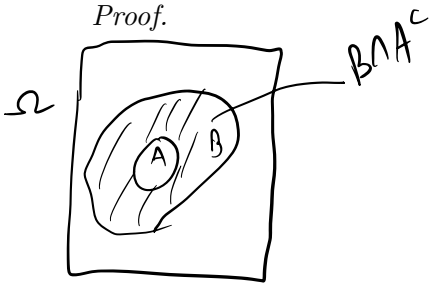
Proof. $A \cup A^c = \Omega$ and $A \cap A^c = \emptyset$ (disjoint)

So $P(\Omega) = P(A) + P(A^c)$

$\Rightarrow \quad 1 = P(A) + P(A^c) \Rightarrow P(A^c) = 1 - P(A)$ □

Theorem 1.8. *Inclusion: If* $A \subset B$, *then* $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Proof.



$B = A \cup (B \cap A^c)$

So $P(B) = P(A) + P(B \cap A^c)$

$\Rightarrow P(B) \geq P(A)$ since $P(B \cap A^c) \geq 0$ □

Theorem 1.9. *Regularity: for every event* A, $\mathbb{P}(A) \in [0, 1]$.

Proof. $P(A) \geq 0$ (axiom 1)

Now $A \subset \Omega$ for any A

So $P(A) \leq P(\Omega) = 1$ (Thm. 1.8) □

Lemma 1.10 (Decomposition identities)**.** *For every two events* A *and* B,

- $\mathbb{P}(A \cap B^C) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$.
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Example 1.13 (Diagnosing diseases)**.** A patient arrives at a doctor's office with a sore throat and low-grade fever. After an exam, the doctor decides that the patient definitely has either a bacterial infection (B) or a viral infection (V) or both. The doctor determines that $\mathbb{P}(B) = 0.7$ and $\mathbb{P}(V) = 0.4$. What is the probability that the disease is due to both B and V?

$\Omega = B \cup V$ we want $P(B \cap V)$

$P(B \cup V) = P(B) + P(V) - P(B \cap V)$

$1 = 0.7 + 0.4 - P(B \cap V)$

$\Rightarrow P(B \cap V) = 0.1$

♢

Lemma 1.11 (Union bound)**.** *For events $A_1, ..., A_n$,*

$$\mathbb{P}(\cup_{i=1}^n A_i) \leq \sum_{i=1}^n \mathbb{P}(A_i).$$

Example 1.14. Suppose you have four classes one day. The probability that you are late to any given class is 0.05. Find an upper bound for the probability that you are late to at least one class. ♢

A_i : event that you are late to class i , $i = 1, ..., 4$

$P(\bigcup_{i=1}^{4} A_i) \leq \sum_{i=1}^{4} P(A_i) = 4 \times 0.05 = \boxed{0.2}$

Lemma 1.12 (Bonferroni inequality:)**.** *For all events $A_1, ..., A_n$,*

$$\mathbb{P}(\cap_{i=1}^n A_i) \geq 1 - \sum_{i=1}^n \mathbb{P}(A_i^C)$$

Example 1.15. Suppose you are taking 3 exams this week. The probability that you pass any given exam is 0.9. Find a lower bound for the probability that you pass all of your exams. ♢

A_i : event that you pass exam i , $i = 1, 2, 3$

$P(A_i) = 0.9 \Rightarrow P(A_i^c) = 1 - 0.9 = 0.1$

$P(\bigcap_{i=1}^{3} A_i) \geq 1 - \sum_{i=1}^{3} P(A_i^c) = 1 - 3 \times 0.1 = 0.7$

1.4 Probability on finite sample spaces

In classical probability theory, sample spaces are often finite—in other words, the number of possible outcomes is finite. For instance, rolling a die, drawing a card from a standard deck, etc..
A general probability theory can be established for the experiments where equally likely outcomes exist.

Definition 1.14 (Equally likely outcomes)**.** Suppose $\Omega = \{\omega_i\}_{i=1}^N$, $\mathbb{P}(\omega_i) = \frac{1}{N}$ for $i = 1, .., N$.

With equally likely outcomes, the probability calculation is simply counting.

Theorem 1.13. *For the finite sample space Ω with equally likely outcomes, the probability of event $E \subset \Omega$ is*

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|},$$

where $|E|$ denotes ***the number of outcomes*** *in E.*

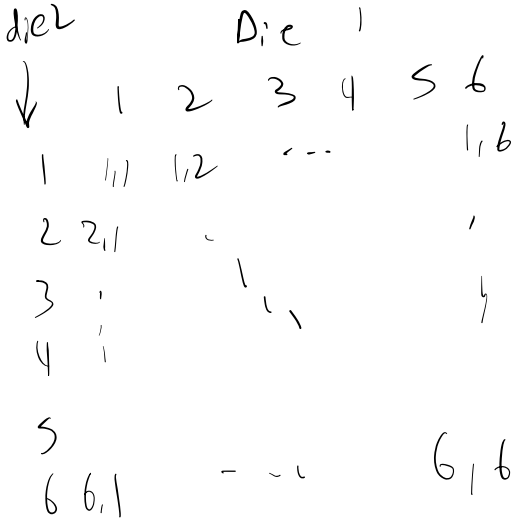
Example 1.16 (Tossing coins)**.** Suppose three fair coins are tossed simultaneously. What is the probability of obtaining exactly two heads?

$E = \{HHT, HTH, THH\}$

$P(E) = \frac{|E|}{|\Omega|} = \boxed{\frac{3}{8}}$

♢

Example 1.17 (Roll two fair dice)**.** Two fair dice are rolled. What are the equally likely outcomes? Which is more likely: sum is 11 or sum is 12?



Event A: 5,6 or 6,5

Event B: 6,6

So $P(A) = \frac{2}{36} > P(B) = \frac{1}{36}$

♢